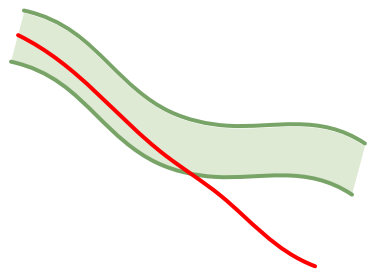
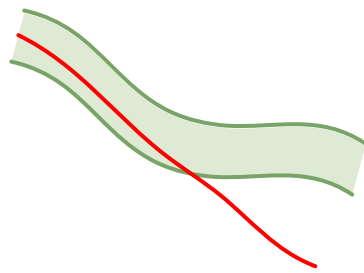


(A) Why Binary-overlap supervision is limited

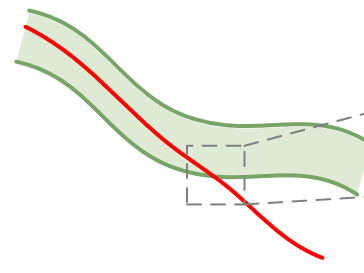
Binary-overlap objective



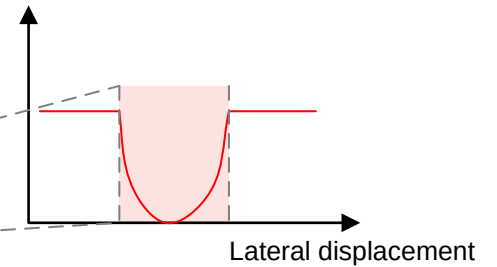
Partial overlap



Limited gradient support

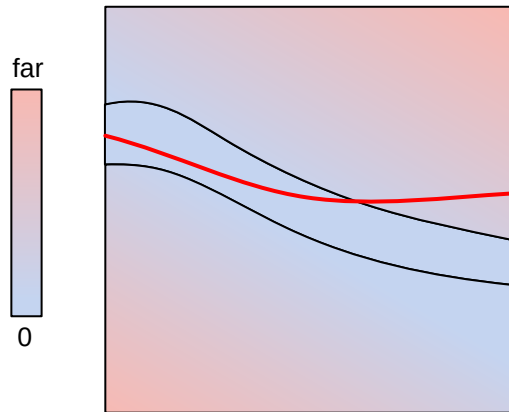


Overlap-based loss

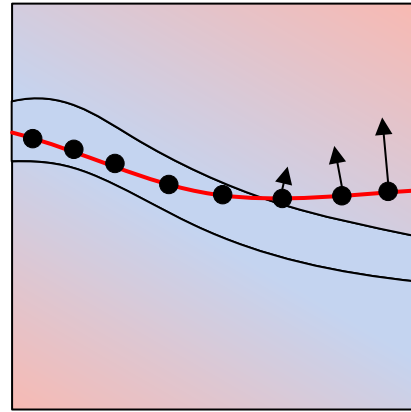


(B) Distance-field containment provides continuous spatial guidance

Distance map

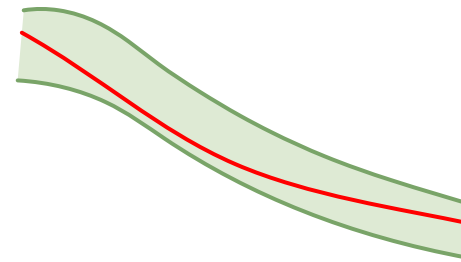


Sample with grid point

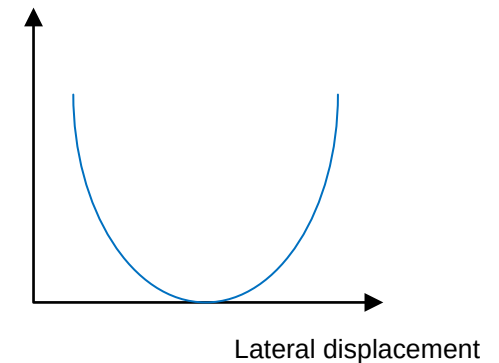


continuous spatial guidance

Guidewire contained



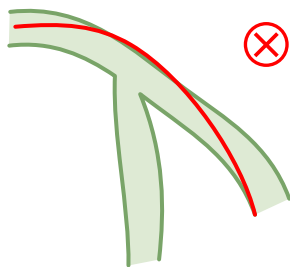
Containment loss



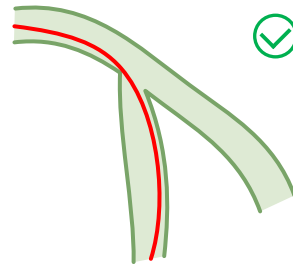
(C) Auxiliary constraints improve robustness

Tangent

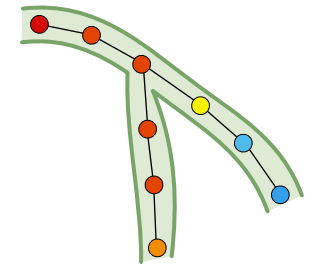
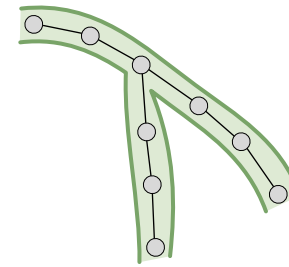
Containment only



+ Tangent consistency



Graph nodes



Vessel lumen



Guidewire



Vessel-lumen distance field



Node weight
(low $\rightarrow$ high)



Graph edge